# Mohammad Javadi Balakan

Research Scientist · San Jose, CA

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## Research interests

Experimental condensed-matter physics of quantum materials and mesoscopic devices: topological superconductivity, flux quantization and phase-coherent transport for fault-tolerant quantum computing; earlier work on 2D-material and hybrid organic–silicon devices for optoelectronics and photovoltaics.

## Appointments

2026 – present **Research Scientist**
Stealth-mode startup, Bay Area, CA

2022 – 2026 **Research Scientist (2025 – 2026); Postdoctoral Fellow (2022 – 2024)**
College of Nanotechnology, Science, and Engineering (CNSE), State University of New York at Albany
Group of Prof. Ji Ung Lee · Topological superconductors for fault-tolerant quantum computation

- Electrical transport studies of topological superconductors in collaboration with Stony Brook University and BNL (Prof. Qiang Li and Prof. Genda Gu).
- Developed processes for end-to-end fabrication of nanoscale quantum devices.
- Upgraded a closed-cycle cryostat, designed and implemented cryogenic low-pass filters (LPFs) for precision electrical measurements, and developed Python scripts for instrument automation.
- Demonstrated half-integer flux quantization, intrinsic magnetization of the superconducting condensate, and Aharonov-Bohm oscillations of edge states in mesoscopic Fe(Te,Se) devices.

2019 – 2021 **Postdoctoral Associate**
Department of Physics, University of Tehran, Iran
Group of Prof. Yaser Abdi · CMOS-compatible solid-state devices based on 2D materials

- Investigated charge transport across graphene-silicon 2D/3D Schottky junctions.
- Developed a model for the fundamental efficiency of graphene-based solar cells.
- Fabricated and characterized infrared detectors based on 2D $MoS_2$ layers.
- Developed scalable position-sensitive detectors based on hybrid organic/inorganic PEDOT:PSS/Silicon heterojunctions (*APL Editor's Pick*).
- Pioneered a scalable method for synthesis of 2D borophene metallic layers using chemical vapor deposition.

Mar. – Sep. 2018 **Visiting Researcher**
Department of Physics, Sapienza Università di Roma, Italy
Supervisor: Prof. Giorgio Parisi · Statistical properties of harmonic spheres near jamming

## Education

2019 **PhD in Solid-State Physics, University of Tehran, Iran**
Thesis: *Charge Transport in Metal-Oxide Nano-Structures for Photovoltaic and Optoelectronic Applications*

2014 **MSc in Nanoscience and Nanotechnology, University of Tehran, Iran**
Thesis: *Electron Transport in Nano-Structured Solar Cells*

2012 **BSc in Physics, Urmia University, Iran**

*Graduate research*

Department of Physics, University of Tehran (Prof. Abdi's and Prof. Arzi's joint group)

- Fabrication and characterization of nanostructured solar cells based on dye-sensitized solar cells (solar simulator, IV, quantum efficiency, carrier lifetime and diffusion length measurement).
- Monte-Carlo simulation of charge transport in disordered wide band-gap semiconductors for photovoltaic and optoelectronic applications.
- Computer modeling for PECVD growth of single-standing carbon nanotubes.

Nanoelectronic laboratory, Tarbiat Modares University, Iran (Dr. Sara Darbari's lab)

- Investigated sound-driven CNT-based tribo-electric nanogenerator in sonic array.
- Developed a piezo-phototronic device based on graphene/MoS2 heterostructure.

## Publications

22 publications and preprints · 436 citations · h-index 13 (Google Scholar, October 2026)

*Accepted*

22. **M. J. Balakan**, S. Heidari, G. Gu, Q. Li, K. Watanabe, T. Taniguchi, and J. U. Lee, "Intrinsic magnetization of the superconducting condensate in Fe(Te,Se)," *Physical Review B* (Letter), accepted (2026). Editors' suggestion arXiv:2603.08891

*Submitted and preprints*

21. **M. J. Balakan**, G. Gu, Q. Li, K. Watanabe, T. Taniguchi, and J. U. Lee, "Half-integer winding of order parameter in an iron-based superconductor," submitted (2026). arXiv:2505.01522
20. **M. J. Balakan**, S. Heidari, G. Gu, Q. Li, D. Titcombe, Z. Xiao, H. Frost, E. Bhatia, C. Kow, P. Rao, K. Watanabe, T. Taniguchi, and J. U. Lee, "Aharonov–Bohm interference from coherent spin-polarized edge transport in Fe(Te,Se) superconducting rings," submitted (2025). arXiv:2508.05030
19. **M. J. Balakan**, G. Gu, Q. Li, K. Watanabe, T. Taniguchi, and J. U. Lee, "Observation of half-quantum vorticity in an iron-based superconductor," preprint (2025). arXiv:2505.01522

*Peer-reviewed journal articles*

18. A. Mazaheri, **M. Javadi**, and Y. Abdi, "Chemical vapor deposition of two-dimensional boron sheets by thermal decomposition of diborane," *ACS Applied Materials & Interfaces* **13**(7), 8844-8850 (2021). doi:10.1021/acsami.0c22580
17. **M. Javadi**, AA. Noroozi, and Y. Abdi, "Kinetics of charge carriers across a graphene–silicon Schottky junction," *Physical Review Applied* **14**, 064048 (2020). doi:10.1103/PhysRevApplied.14.064048

16. AH. Mosaddegh, AA. Noroozi, **M. Javadi**, and Y. Abdi, "Ultrahigh photovoltage responsivity of PEDOT:PSS–silicon hybrid heterojunction photodiodes," *Applied Physics Letters* **117**, 073301 (2020). doi:10.1063/5.0007685
15. **M. Javadi**, "Theoretical efficiency limit of graphene–semiconductor solar cells," *Applied Physics Letters* **117**, 053902 (2020). doi:10.1063/5.0020080
14. **M. Javadi**, A. Mazaheri, H. Torbatiyan, Y. Abdi, "Mechanism of charge transport in hybrid organic–inorganic PEDOT:PSS/silicon heterojunctions," *Physical Review Applied* **12**, 034002 (2019). doi:10.1103/PhysRevApplied.12.034002
13. **M. Javadi**, AA. Noroozi, A. Mazaheri, Y. Abdi, "Sequentially assembled graphene layers on silicon, the role of uncertainty principles in graphene–silicon Schottky junctions," *Advanced Optical Materials* **7**, 1900470 (2019). doi:10.1002/adom.201900470
12. **M. Javadi**, Y. Abdi, "Frequency-driven bulk-to-surface transition of conductivity in ZnO nanowires," *Applied Physics Letters* **113**, 051603 (2018). doi:10.1063/1.5039474
11. **M. Javadi**, M. Gholami, H. Torbatiyan, Y. Abdi, "Hybrid organic/inorganic position-sensitive detectors based on PEDOT:PSS/n-Si," *Applied Physics Letters* **112**, 113302 (2018). Editor's Pick doi:10.1063/1.5022758
10. **M. Javadi**, M. Gholami, Y. Abdi, "IR position-sensitive detectors based on double-junction asymmetric $TiO_2$/$MoS_2$/reduced graphene-oxide sandwiches," *Journal of Materials Chemistry C* **6**, 8444 (2018). doi:10.1039/c8tc02361a
9. **M. Javadi**, A. Heidari, S. Darbari, "Realization of enhanced sound-driven CNT-based triboelectric nanogenerator, utilizing sonic array configuration," *Current Applied Physics* **18**, 361 (2018). doi:10.1016/j.cap.2018.01.018
8. **M. Javadi**, H. Torbatiyan, Y. Abdi, "Ultra-low intensity UV detection using partitioned mesoporous $TiO_2$," *Applied Physics Letters* **111**, 063113 (2017). doi:10.1063/1.4994888
7. **M. Javadi**, Y. Abdi, "Kinetic Monte Carlo simulation of single-electron multiple-trapping transport in disordered media," *Computer Physics Communications* **221**, 282 (2017). doi:10.1016/j.cpc.2017.09.002
6. **M. Javadi**, Y. Abdi, "Nanopattern formation using localized plasma for growth of single-standing carbon nanotubes," *Journal of Nanoparticle Research* **19**, 17 (2017). doi:10.1007/s11051-016-3715-4
5. **M. Javadi**, S. Alizadeh, Y. Khosravi, Y. Abdi, "Electron transport in Q2D porous network of titania nanoparticles, incorporating electrical and optical advantages in dye-sensitized solar cells," *ChemPhysChem* **17**, 3542 (2016). doi:10.1002/cphc.201600609
4. **M. Javadi**, S. Darbari, Y. Abdi, F. Ghasemi, "Realization of a piezo-phototronic device based on reduced graphene-oxide/$MoS_2$ heterostructure," *IEEE Electron Device Letters* **37**, 677 (2016). doi:10.1109/LED.2016.2547993
3. **M. Javadi**, Y. Abdi, E. Arzi, "Local collection efficiency in nano-crystalline solar cells," *Solar Energy* **133**, 549 (2016). doi:10.1016/j.solener.2016.04.021
2. M. Sajedi Alvar, **M. Javadi**, Y. Abdi, E. Arzi, "Enhancing the electron lifetime and diffusion coefficient in dye-sensitized solar cells by patterning the layer of titanium dioxide nanoparticles," *Journal of Applied Physics* **119**, 114302 (2016). doi:10.1063/1.4943772

1. **M. Javadi**, and Y. Abdi, "Monte Carlo random walk simulation of electron transport in confined porous $TiO_2$ as a promising candidate for photo-electrode of nano-crystalline solar cells," *Journal of Applied Physics* **118**, 064304 (2015). doi:10.1063/1.4928328

## Invited talks

- *"Half-integer winding of order parameter & intrinsic magnetization of superconducting condensate in Fe(Te,Se)"*, GRC correlated electron systems, Mount Holyoke College, MA, USA (2026).
- "*Topological superconductivity in Fe(Te,Se)*", Laboratory for Physical Sciences (LPS), College Park, MD, USA (2025).
- "*Charge transport at graphene-silicon 2D/3D interface*", *TMU*, Tehran, Iran (2020).
- "*Negative photoconductivity in ZnO nanowires*", IPM, Tehran, Iran (2018).

## Conference presentations

- **M. J. Balakan**, Half-integer winding of the order parameter and intrinsic magnetization of the superconducting condensate in mesoscopic Fe(Te,Se) devices, GRC Correlated Electron Systems (2026).
- **M. J. Balakan**, G. Gu, Q. Li, K. Watanabe, T. Taniguchi, and JU. Lee, Topological superconductivity in Fe(Te,Se) mesoscopic devices, APS Global Physics Summit (2026).
- Additional presentations at Iranian Physical Society meetings and nanostructured solar cell conferences (2014–2018).

## Awards, fellowships and funding

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| 2020 | Chamran Postdoctoral Fellowship, Iran National Elites Foundation (INEF) |
| 2019 | Outstanding PhD Thesis in Physics (nationwide), Physical Society of Iran |
| 2019 | Outstanding PhD Dissertation in Physical Sciences, University of Tehran |
| 2016–2018 | INEF Graduate Scholarship |
| 2014 | Ranked 10th nationally in the PhD entrance examination |
| 2014 | Graduated with distinction, MSc in Physics |
| 2012 | Ranked 19th nationally in the MSc entrance examination |
| 2012 | Graduated with distinction, BSc in Physics |

## Teaching and mentoring

Teaching Assistant, University of Tehran (2012–2018): Quantum Mechanics (undergraduate), Advanced Solid-State Physics & Laboratory (graduate), Advanced Quantum Mechanics (graduate)

- Mentored Ph.D. researcher Daniel Titcomb in experimental studies of topological superconductors (Dr. Lee's group).
- Mentored graduate researcher AliAkbar Noroozi in graphene-silicon optical detectors; work contributed to a Ph.D. thesis and two publications (Dr. Abdi's group).

- Mentored graduate student Mahdieh Gholami on CVD growth of two-dimensional $MoS_2$ for infrared detectors; work contributed to a master's thesis and a publication (Dr. Abdi's group).

## Professional service

- Peer review for scientific journals including Physical Review Letters, 2D Materials, Applied Physics Letters, and Nanotechnology (ORCID).
- Served as a committee member for the Physical Society of Iran symposium, 2018.

## Technical skills

**Fabrication:** cleanroom nanofabrication and process development; EBL, FIB, RIE; PVD, e-beam evaporation, CVD, PECVD; 2D materials and hBN encapsulation.

**Measurement:** cryogenic quantum transport; low-noise lock-in and phase-sensitive flux measurements; cryogenic filter design; SEM, AFM, Raman; optoelectronic characterization.

**Computation:** Python, Mathematica, LabVIEW, KLayout; instrument automation; Monte Carlo and COMSOL modeling.

## References

Available upon request.